

Elementary Rotations and Rigid-Body Dynamics — approved segments

48 approved segments, in reading order

SUMMARY ON ELEMENTARY ROTATIONS:

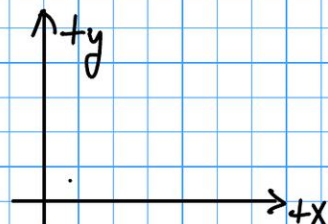
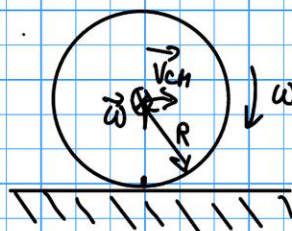
$$L_z = I \omega_z \quad (1) \quad \vec{N} = \vec{r} \times \vec{F} \quad (2) \quad T = \frac{I}{2} \omega_z^2 = \frac{\omega_z L_z}{2} \quad (3)$$

$$\frac{d\vec{L}}{dt} = \vec{N}(\vec{e}) \quad (4) \quad \left\{ \begin{array}{l} \text{INERTIAL FRAME} \\ \text{CM FRAME (ORIGIN AT CM,} \\ \text{AXES PARALLEL TO INERTIAL AXES)} \end{array} \right.$$

Rolling without
slipping condition:

$$v_{Cx} + \omega_z R = 0 \quad (5)$$

($\omega_z < 0$ for clockwise rotation)
(6)



Parallel Axis Theorem:

$$I = \bar{I} + M a^2 \quad (7)$$

AXIS

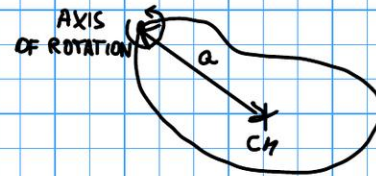
(6)

Parallel Axis Theorem:

$$I = \bar{I} + Ma^2 \quad (7)$$

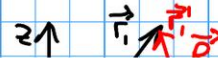
(8) $\bar{I} \equiv$ Moment of inertia around CM

(9) $a \equiv$ Distance between axis of rotation and CM



Separating CM Motion:

$$M = \sum m_i \quad (12)$$



(9) $a \equiv$ Distance between axis of rotation and CM

Separating CM Motion:

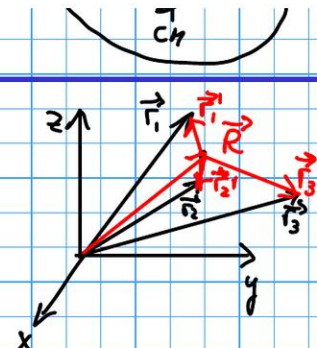
$$M = \sum m_j \quad (12)$$

$$\vec{L} = \vec{R} \times M \dot{\vec{R}} + \sum_j \vec{r}_j \times m_j \dot{\vec{r}}_j \quad (10)$$

$$\vec{R} = \frac{\sum m_j \vec{r}_j}{M} \quad (13)$$

$$T = \frac{M}{2} |\dot{\vec{R}}|^2 + \frac{1}{2} \sum_j m_j |\dot{\vec{r}}_j|^2 \quad (11)$$

$$\vec{r}_j = \vec{r}_j - \vec{R} \quad (14) \quad \text{POSITION RELATIVE TO CM}$$



For rigid body, considering rotation with angular velocity ω_z around CM:

$$L_z = (\vec{R} \times M \dot{\vec{R}})_z + I \omega_z \quad (16)$$

$$T = \frac{M}{2} |\dot{\vec{R}}|^2 + \frac{1}{2} I \omega_z^2 \quad (17)$$

PROOFS:

$$\vec{L} = \sum_j \vec{r}_j \times m_j \dot{\vec{r}}_j = \sum_j (\vec{R} + \vec{r}_j') \times m_j \dot{\vec{r}}_j = \vec{R} \times M \dot{\vec{R}} + \sum_j \vec{r}_j' \times m_j \dot{\vec{r}}_j \quad (18)$$

$$L_z = (\mathbf{R} \times M \dot{\mathbf{R}})_z + \sum \omega_z \quad (16)$$

$$I = \frac{1}{2} |\mathbf{R}|^2 + \frac{1}{2} \omega_z^2 \quad (17)$$

PROOFS:

CHANGING TO
CM SYSTEM;

$$\mathbf{L} = \sum_{i=1}^N \mathbf{r}_i \times m_i \dot{\mathbf{r}}_i = \sum_{i=1}^N (\mathbf{R} + \mathbf{r}'_i) \times m_i (\dot{\mathbf{R}} + \dot{\mathbf{r}}'_i) = \sum_i \mathbf{R} \times m_i \dot{\mathbf{R}} + \sum_i \mathbf{r}'_i \times m_i \dot{\mathbf{r}}'_i + \text{CROSS-TERMS} \quad (18)$$

$$\mathbf{r}_i = \mathbf{R} + \mathbf{r}'_i \quad \dot{\mathbf{r}}_i = \dot{\mathbf{R}} + \dot{\mathbf{r}}'_i \quad (19)$$

$$\text{CROSS-TERMS} \left\{ \begin{aligned} \sum_i \mathbf{r}'_i \times m_i \dot{\mathbf{R}} &= \sum_i m_i (\mathbf{r}'_i - \dot{\mathbf{R}}) \times \dot{\mathbf{R}} = \left[\sum_i m_i \mathbf{r}'_i - \left(\sum_i m_i \right) \dot{\mathbf{R}} \right] \times \dot{\mathbf{R}} = [M \dot{\mathbf{R}} - M \dot{\mathbf{R}}] \times \dot{\mathbf{R}} = \mathbf{0} \quad (20) \\ \sum_i \mathbf{R} \times m_i \dot{\mathbf{r}}'_i &= \mathbf{R} \times \sum_i m_i (\dot{\mathbf{r}}'_i - \dot{\mathbf{R}}) = \mathbf{R} \times \left[\sum_i m_i \dot{\mathbf{r}}'_i - \left(\sum_i m_i \right) \dot{\mathbf{R}} \right] = \mathbf{R} \times [M \dot{\mathbf{R}} - M \dot{\mathbf{R}}] = \mathbf{0} \quad (21) \end{aligned} \right.$$

$$\text{THEN: } \mathbf{L} = \mathbf{R} \times M \dot{\mathbf{R}} + \sum_i \mathbf{r}'_i \times m_i \dot{\mathbf{r}}'_i \quad (22) \Rightarrow \frac{d\mathbf{L}}{dt} = \mathbf{R} \times M \ddot{\mathbf{R}} + \sum_i \mathbf{r}'_i \times m_i \ddot{\mathbf{r}}'_i \quad (23) \quad \text{TIME DERIVATIVE}$$

$$\text{BUT ALSO: } \frac{d\mathbf{L}}{dt} = \mathbf{N}^{(e)} = \sum_i \mathbf{r}_i \times \mathbf{F}_i^{(e)} \quad (24)$$

$$\frac{d\mathbf{L}}{dt} = \mathbf{R} \times \sum_i \mathbf{F}_i^{(e)} + \sum_i \mathbf{r}'_i \times \mathbf{F}_i^{(e)} = \mathbf{R} \times M \ddot{\mathbf{R}} + \sum_i \mathbf{r}'_i \times \mathbf{F}_i^{(e)} \quad (26)$$

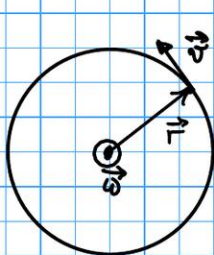
(25) $M \ddot{\mathbf{R}} = \sum_i \mathbf{F}_i^{(e)}$ Newton's 2nd law for particle system

$$\frac{d\mathbf{L}'}{dt} = \sum_{i=1}^N \mathbf{r}'_i \times \mathbf{F}_i^{(e)} \quad (27)$$

Eq (27) is the special case of Eq (4) in the CM Frame.

Rotations, Non-inertial systems, and Rigid body motion

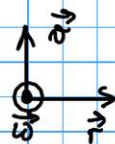
CIRCULAR MOTION
IN A PLANE



$$\vec{v} = \vec{\omega} \times \vec{r}$$

$$\vec{v} \perp \vec{r}$$

$$\vec{r} \cdot \vec{r} = r^2 = 0$$



$$\hat{e}_i \cdot \hat{e}_j = \delta_{ij} = \begin{cases} 1 & i=j \\ 0 & i \neq j \end{cases}$$

ROTATING COORDINATE SYSTEM



$$0 = \frac{d\delta_{ij}}{dt} = \frac{d\hat{e}_i \cdot \hat{e}_j}{dt}$$

$$0 = \hat{e}_i \cdot \frac{d\hat{e}_j}{dt} \quad i=j$$

$$0 = \hat{e}_i \cdot \frac{d\hat{e}_j}{dt} + \hat{e}_j \cdot \frac{d\hat{e}_i}{dt} \quad i \neq j$$

$(\hat{e}_j)_{j=1,2,3}$ IS A BASIS $\Rightarrow$ WE CAN WRITE $\frac{d\hat{e}_i}{dt}$ AS A LINEAR COMBINATION

$$\frac{d\hat{e}_i}{dt} = \sum_{j=1}^3 \Omega_{ij} \hat{e}_j \quad \Omega_{ij} = \hat{e}_j \cdot \frac{d\hat{e}_i}{dt}$$

(USE THAT $\hat{e}_j \cdot \sum_{i=1}^3 \Omega_{ij} \hat{e}_i = \sum_{i=1}^3 \Omega_{ij} \hat{e}_j \cdot \hat{e}_i = \Omega_{ij}$)

$$\hat{x} \times \hat{y} = \hat{z}$$

$$\hat{y} \times \hat{z} = \hat{x}$$

$$\hat{z} \times \hat{x} = \hat{y}$$

$$\hat{e}_1 \equiv \hat{x} \quad \hat{e}_2 \equiv \hat{y} \quad \hat{e}_3 \equiv \hat{z}$$

$$\hat{e}_1 \times \hat{e}_2 = \hat{e}_3 \quad (\text{AND CYCLICALLY})$$

$$\hat{e}_2 \times \hat{e}_3 = \hat{e}_1$$

$$\hat{e}_3 \times \hat{e}_1 = \hat{e}_2$$

Levi-Civita symbol

$$\epsilon_{ijk} = \begin{cases} +1 & ijk \text{ cyc } 123 \\ -1 & ijk \text{ cyc } 321 \\ 0 & \text{if any repeated index} \end{cases}$$

$$\epsilon_{123} = \epsilon_{231} = \epsilon_{312} = +1$$

$$\epsilon_{213} = \epsilon_{132} = \epsilon_{321} = -1$$

$$\epsilon_{iij} = \epsilon_{ijj} = \epsilon_{jij} = 0$$

$$\hat{e}_i \times \hat{e}_j = \sum_k \epsilon_{ijk} \hat{e}_k$$

$$\langle \varphi_i | \left(\sum_j c_j |\varphi_j\rangle \right) = \sum_j c_j \langle \varphi_i | \varphi_j \rangle = c_i$$

$$i \neq j \quad 0 = \Omega_{ij} + \Omega_{ji} \Rightarrow \Omega_{ji} = -\Omega_{ij} \quad \begin{pmatrix} 0 & \Omega_{12} & -\Omega_{31} \\ -\Omega_{12} & 0 & \Omega_{23} \\ \Omega_{31} & -\Omega_{23} & 0 \end{pmatrix} \quad \omega_1 = \Omega_{23} \quad \omega_2 = \Omega_{31} \quad \omega_3 = \Omega_{12}$$

$$\Omega_{ii} = \hat{e}_i \cdot \frac{d\hat{e}_i}{dt} = 0$$

$$\frac{d\hat{e}_1}{dt} = \Omega_{12} \hat{e}_2 + \Omega_{13} \hat{e}_3 = \Omega_{12} \hat{e}_2 - \Omega_{31} \hat{e}_3 = \omega_3 \hat{e}_2 - \omega_2 \hat{e}_3 = \vec{\omega} \times \hat{e}_1 \quad \checkmark$$

$$\vec{\omega} \times \hat{e}_1 = (\omega_1 \hat{e}_1 + \omega_2 \hat{e}_2 + \omega_3 \hat{e}_3) \times \hat{e}_1 = -\omega_2 \hat{e}_3 + \omega_3 \hat{e}_2$$

$$\frac{d\hat{e}_2}{dt} = \Omega_{21} \hat{e}_1 + \Omega_{23} \hat{e}_3 = -\Omega_{12} \hat{e}_1 + \Omega_{23} \hat{e}_3 = -\omega_3 \hat{e}_1 + \omega_1 \hat{e}_3 = \vec{\omega} \times \hat{e}_2 \quad \checkmark$$

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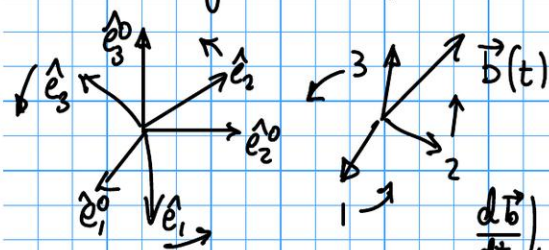
$$\frac{d\hat{e}_2}{dt} = \Omega_{21} \hat{e}_1 + \Omega_{23} \hat{e}_3 = -\Omega_{12} \hat{e}_1 + \Omega_{32} \hat{e}_3 = -\omega_3 \hat{e}_1 + \omega_1 \hat{e}_3 = \vec{\omega} \times \hat{e}_2 \quad \checkmark$$

$$\vec{\omega} \times \hat{e}_2 = (\omega_1 \hat{e}_1 + \omega_2 \hat{e}_2 + \omega_3 \hat{e}_3) \times \hat{e}_2 = \omega_1 \hat{e}_3 - \omega_3 \hat{e}_1$$

$\hat{e}_1 \times \hat{e}_2 = \hat{e}_3 \quad \hat{e}_3 \times \hat{e}_2 = -\hat{e}_1$

$$\frac{d\hat{e}_3}{dt} = \vec{\omega} \times \hat{e}_3$$

Rotating reference frames



Fixed inertial reference frame $\hat{e}_1^0, \hat{e}_2^0, \hat{e}_3^0$

Rotating (non-inertial) reference frame $\hat{e}_1^i, \hat{e}_2^i, \hat{e}_3^i$

$$\left. \frac{d\vec{b}}{dt} \right|_{\text{inertial}} = \frac{d}{dt} \left(\sum_{i=1}^3 b_i^0 \hat{e}_i^0 \right) = \sum_{i=1}^3 \dot{b}_i^0 \hat{e}_i^0$$

$$\left. \frac{d\vec{b}}{dt} \right|_{\text{inertial}} = \frac{d}{dt} \left(\sum_{i=1}^3 b_i \hat{e}_i \right) = \sum_{i=1}^3 \dot{b}_i \hat{e}_i + \sum_{i=1}^3 b_i \dot{\hat{e}}_i \quad \left. \frac{d\vec{b}}{dt} \right|_{\text{body}} = \sum_{i=1}^3 \frac{db_i}{dt} \hat{e}_i$$

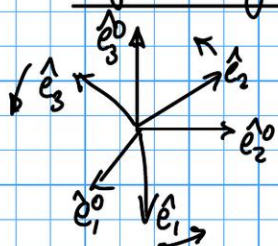
$$\dot{\hat{e}}_i = \frac{d\hat{e}_i}{dt} = \vec{\omega} \times \hat{e}_i$$

For rotations $\left. \frac{d\vec{b}}{dt} \right|_{\text{inertial}} = \left. \frac{d\vec{b}}{dt} \right|_{\text{body}} + \sum_{i=1}^3 b_i (\vec{\omega} \times \hat{e}_i)$

$$\boxed{\left. \frac{d\vec{b}}{dt} \right|_{\text{inertial}} = \left. \frac{d\vec{b}}{dt} \right|_{\text{body}} + \vec{\omega} \times \vec{b}}$$

Displacement

Rigid body rotations



$\vec{r}_p$: position of point p in the body frame (CONSTANT IN BODY FRAME)
Rigid body with one fixed point.

$$T = \sum_p \frac{1}{2} m_p v_p^2 = \sum_p \frac{1}{2} m_p (\vec{\omega} \times \vec{r}_p)^2$$

Levi-Civita symbol

$$\epsilon_{ijk} = \begin{cases} +1 & \text{ijk cyc 123} \\ -1 & \text{ijk cyc 213} \\ 0 & \text{if any repeated index} \end{cases}$$

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$$\epsilon_{213} = \epsilon_{132} = \epsilon_{321} = -1$$

$$\epsilon_{dii} = \epsilon_{ijj} = \epsilon_{jij} = 0$$

$$\hat{e}_1 \times \hat{e}_2 = \hat{e}_3 = \sum_l \epsilon_{12l} \hat{e}_l \checkmark$$

$$\sum_l \epsilon_{12l} \hat{e}_l = \hat{e}_3$$

$$\hat{e}_i \times \hat{e}_j = \sum_l \epsilon_{ijl} \hat{e}_l \left(\underset{\text{NOTATION}}{=} \epsilon_{ijl} \hat{e}_l \right)$$

$$\vec{a} \times \vec{b} = \left(\sum_i a_i \hat{e}_i \right) \times \left(\sum_j b_j \hat{e}_j \right) = \sum_{ijl} a_i b_j \epsilon_{ijl} \hat{e}_l = \sum_{ijk} \epsilon_{ijk} a_i b_j \hat{e}_k \quad (\vec{a} \times \vec{b})_k = \sum_{ij} \epsilon_{ijk} a_i b_j$$

Also

$$\sum_i \epsilon_{jki} \epsilon_{lmk} = \sum_i \epsilon_{ijk} \epsilon_{ilm} = \delta_{jl} \delta_{km} - \delta_{jm} \delta_{kl}$$

$$j \xrightarrow{k} i \quad \epsilon_{ijk} = \epsilon_{jki} = \epsilon_{kij} \quad (\text{CYCLIC})$$

$$\sum_i \epsilon_{jki} \epsilon_{lmk} = \sum_i \epsilon_{ijk} \epsilon_{ilm} = \delta_{jl} \delta_{km} - \delta_{jm} \delta_{lk}$$

$$\begin{matrix} & i \\ & \uparrow \\ j & \circlearrowleft & k \end{matrix} \quad \epsilon_{ijk} = \epsilon_{jki} = \epsilon_{kij} \quad (\text{cyclic})$$

$$\begin{aligned} (\vec{A} \times \vec{B}) \cdot (\vec{C} \times \vec{D}) &= \left(\sum_{ijk} A_i B_j \epsilon_{ijk} \hat{e}_k \right) \cdot \left(\sum_{lmn} C_l D_m \epsilon_{lmn} \hat{e}_n \right) = \sum_{ijklm} \epsilon_{ijk} \epsilon_{lmk} A_i B_j C_l D_m \\ &= \sum_{ijklm} (\delta_{il} \delta_{jm} - \delta_{im} \delta_{jl}) A_i B_j C_l D_m = (\vec{A} \cdot \vec{C})(\vec{B} \cdot \vec{D}) - (\vec{A} \cdot \vec{D})(\vec{B} \cdot \vec{C}) \end{aligned}$$

IN SUMMARY:

$$T = \sum_P \frac{1}{2} m_P v_P^2 = \sum_P \frac{1}{2} m_P (\vec{\omega} \times \vec{r}_P)^2 = \sum_P \frac{m_P}{2} [\omega^2 r_P^2 - (\vec{\omega} \cdot \vec{r}_P)^2]$$

$$(\vec{A} \times \vec{B}) \cdot (\vec{C} \times \vec{D}) = (\vec{A} \cdot \vec{C})(\vec{B} \cdot \vec{D}) - (\vec{A} \cdot \vec{D})(\vec{B} \cdot \vec{C}) \Rightarrow (\vec{\omega} \times \vec{r}_P) \cdot (\vec{\omega} \times \vec{r}_P) = (\vec{\omega} \cdot \vec{\omega})(\vec{r}_P \cdot \vec{r}_P) - (\vec{\omega} \cdot \vec{r}_P)(\vec{\omega} \cdot \vec{r}_P)$$

$$T = \frac{1}{2} \sum m_i r_i^2 \omega^2 - \frac{1}{2} \sum m_i (\vec{\omega} \cdot \vec{r}_i)^2 = \frac{1}{2} \sum m_i r_i^2 \omega^2 - \frac{1}{2} \sum m_i (\omega^2 r_i^2 \cos^2 \theta_i)$$

$$(A \times B) \cdot (C \times D) = (A \cdot C)(B \cdot D) - (A \cdot D)(B \cdot C) \Rightarrow (\vec{\omega} \times \vec{r}_P) \cdot (\vec{\omega} \times \vec{r}_P) = (\vec{\omega} \cdot \vec{\omega})(\vec{r}_P \cdot \vec{r}_P) - (\vec{\omega} \cdot \vec{r}_P)(\vec{\omega} \cdot \vec{r}_P)$$

$$T = \frac{1}{2} \sum_P m_P \left[r_P^2 \sum_{i=1}^3 \omega_i \omega_i - \left(\sum_{i=1}^3 \omega_i r_{Pi} \right) \left(\sum_{j=1}^3 \omega_j r_{Pj} \right) \right] = \frac{1}{2} \sum_P m_P \sum_{i,j=1}^3 \left[r_P^2 \delta_{ij} - r_{Pi} r_{Pj} \right] \omega_i \omega_j$$

$$T = \frac{1}{2} \sum_{i,j=1}^3 I_{ij} \omega_i \omega_j$$

$$I_{ij} \equiv \sum_P m_P \left[r_P^2 \delta_{ij} - r_{Pi} r_{Pj} \right]$$

Moment of inertia
tensor (SYMMETRIC)

$$\vec{r} = r \hat{r} \Rightarrow \vec{v} = \frac{d\vec{r}}{dt} = \frac{dr}{dt} \hat{r} + r \frac{d\hat{r}}{dt} \Rightarrow \sqrt{\dot{r}^2 + r^2 \dot{\theta}^2}$$

$$\vec{L} = \sum_p m_p \vec{r}_p \times \vec{v}_p = \sum_p m_p \vec{r}_p \times (\vec{\omega} \times \vec{r}_p)$$

Aside: Using summation notation:

$$L = \sum_p m_p \vec{r}_p \times \vec{v}_p = \sum_p m_p \vec{r}_p \times (\vec{\omega} \times \vec{r}_p)$$

Aside: Using summation notation:

$$\begin{aligned} [A \times (B \times C)]_i &= \epsilon_{ijk} A_j \epsilon_{klm} B_l C_m = \epsilon_{kij} \epsilon_{klm} A_j B_l C_m = (\delta_{il} \delta_{jm} - \delta_{im} \delta_{jl}) A_j B_l C_m \\ &= (\vec{A} \cdot \vec{C}) B_i - (\vec{A} \cdot \vec{B}) C_i \end{aligned}$$

$$[\vec{r}_p \times (\vec{\omega} \times \vec{r}_p)]_i = (\vec{r}_p \cdot \vec{r}_p) \omega_i - (\vec{r}_p \cdot \vec{\omega}) r_{pi}$$

$$L_i = \sum_p m_p (\vec{r}_p \times \vec{v}_p)_i = \sum_p m_p [\vec{r}_p \times (\vec{\omega} \times \vec{r}_p)]_i = \sum_p m_p \left[r_p^2 \omega_i - \sum_{j=1}^3 r_{pi} r_{pj} \omega_j \right]$$

$$L_i = \sum_p \left(\sum_{j=1}^3 \left(\delta_{im} r_{pj}^2 \delta_{jj} - r_{pi} r_{pj} \right) \right) \omega_j = \sum_p \tau_{ij} \omega_j$$

$$L_i = \sum_p m_p (\mathbf{r}_p \times \mathbf{v}_p)_i = \sum_p m_p (\mathbf{r}_p \times (\boldsymbol{\omega} \times \mathbf{r}_p))_i = \sum_p m_p (\mathbf{r}_p \boldsymbol{\omega}_i - \sum_{j=1}^3 r_{pi} r_{pj} \omega_j)$$

$$L_i = \sum_{j=1}^3 \underbrace{\left(\sum_p m_p [\mathbf{r}_p^2 \delta_{ij} - r_{pi} r_{pj}] \right)}_{= I_{ij}} \omega_j = \sum_{j=1}^3 I_{ij} \omega_j$$

$$L_i = \sum_j I_{ij} \omega_j$$

$$T = \frac{1}{2} \boldsymbol{\omega} \cdot \mathbf{L}$$

$$T = \frac{1}{2} \sum_{i,j=1}^3 I_{ij} \omega_i \omega_j$$

$$I_{ij} = I_{ji} = I_{ji}^* \Rightarrow \text{HERMITIAN} \Rightarrow \text{CAN BE DIAGONALIZED} \quad \begin{pmatrix} I_1 & 0 & 0 \\ 0 & I_2 & 0 \\ 0 & 0 & I_3 \end{pmatrix}$$

Directions that diagonalize I_{ij} are called **PRINCIPAL AXES**

EXAMPLES

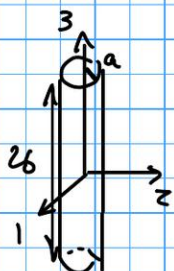
EXAMPLES

Homogeneous CUBE OR SPHERE

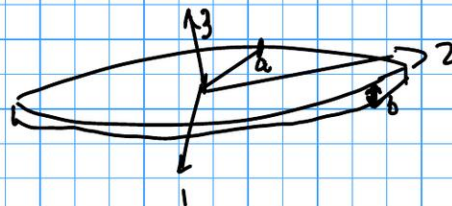


$$\begin{pmatrix} I & 0 & 0 \\ 0 & I & 0 \\ 0 & 0 & I \end{pmatrix} = I_{3 \times 3}$$

$$a \gg b$$



$$a \ll b \Rightarrow I_1 = I_2 > I_3$$



$$a \gg b \Rightarrow I_1 = I_2 < I_3$$

$$I = \sum_p m_p (\text{distance to axis})_p^2$$

$$= \sum_p m_p r_{p1}^2$$

$$I_3 \sim \# M a^2 \quad I_1 \sim \# M b^2$$

$$a \ll b \Rightarrow I_3 \ll I_1$$

$$a \gg b \Rightarrow I_3 \gg I_1$$

$$I_{ij} = \sum_p m_p [\delta_{ij} r_p^2 - r_{pi} r_{pj}]$$

$$I_{33} = \sum_p m_p [\delta_{33} r_p^2 - r_{p3} r_{p3}]$$

$$r_{p1} r_{p1} + r_{p2} r_{p2} + r_{p3} r_{p3} - r_{p3} r_{p3} = r_{p1}^2 + r_{p2}^2 = x^2 + y^2 = \rho^2$$

$$I_3 \sim \frac{1}{2} M a^2 \quad I_1 \sim \frac{1}{2} M b^2$$

$$x^2 + y^2 + z^2 - z^2 = x^2 + y^2 = \rho^2$$

$$a \ll b \Rightarrow I_3 \ll I_1$$

$$a \gg b \Rightarrow I_3 \gg I_1$$

Note irregular: I_1, I_2, I_3 all different

$$\begin{pmatrix} I_1 & 0 & 0 \\ 0 & I_2 & 0 \\ 0 & 0 & I_3 \end{pmatrix} \Leftrightarrow I_{ij} = I_i \delta_{ij} \quad L_i = \sum_{j=1}^3 I_{ij} \omega_j = \sum_{j=1}^3 I_i \delta_{ij} \omega_j = I_i \omega_i$$

$$\begin{pmatrix} L_1 \\ L_2 \\ L_3 \end{pmatrix} = \begin{pmatrix} I_1 & 0 & 0 \\ 0 & I_2 & 0 \\ 0 & 0 & I_3 \end{pmatrix} \begin{pmatrix} \omega_1 \\ \omega_2 \\ \omega_3 \end{pmatrix} = \begin{pmatrix} I_1 \omega_1 \\ I_2 \omega_2 \\ I_3 \omega_3 \end{pmatrix}$$

$$T = \frac{1}{2} \vec{\omega} \cdot \vec{L} = \frac{1}{2} \sum_i \omega_i L_i = \frac{1}{2} \sum_i \omega_i I_i \omega_i = \frac{1}{2} \sum_i I_i \omega_i^2$$

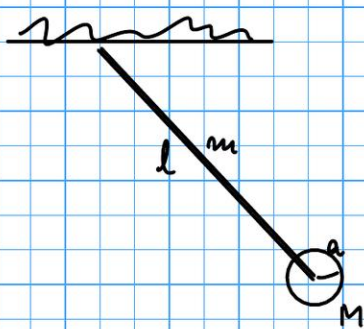
$$(L_i) \quad (0 \ 0 \ I_3) \quad (\omega_i) \quad (I_3 \omega_i)$$

$$T = \frac{1}{2} \vec{\omega} \cdot \vec{L} = \frac{1}{2} \sum_{i=1}^3 \omega_i L_i = \frac{1}{2} \sum_{i=1}^3 \omega_i I_i \omega_i = \frac{1}{2} \sum_{i=1}^3 I_i \omega_i^2$$

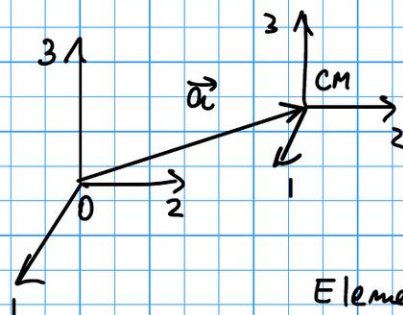
$$L_i = I_i \omega_i$$

$$T = \frac{1}{2} \sum_{i=1}^3 I_i \omega_i^2$$

Parallel Axis Theorem



($\bar{I}_{ij}$ MoI wrt CM)



$$I_{ij} = \bar{I}_{ij} + M[s_{ij}a^2 - a_i a_j]$$

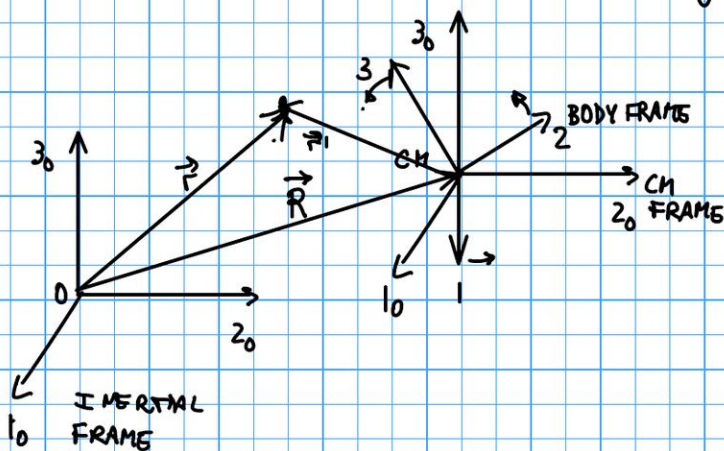
$\updownarrow$ ORIGIN O $\updownarrow$ ORIGIN CM

Elementary version: $I = \bar{I} + M d^2$

Rotation wrt a different origin point

Rotation wrt an arbitrarily moving point

No longer look at a point in the body fixed in space



- INERTIAL FRAME

- CH FRAME { MAY NOT BE INERTIAL IF $\ddot{\vec{R}} \neq \vec{0}$
AXES // AXES OF INERTIAL FRAME

- BODY FRAME { ORIGIN AT $\vec{R} \Rightarrow$ NON-INERTIAL $\ddot{\vec{R}} \neq \vec{0}$
AXES 123 ROTATING WRT $1_0 2_0 3_0$

$$\vec{r} = \vec{R} + \vec{r}'$$

$$\left. \frac{d\vec{r}}{dt} \right|_{\text{INERTIAL}} = \frac{d\vec{R}}{dt} + \left. \frac{d\vec{r}'}{dt} \right|_{\text{CH}}$$

$$\left. \frac{d\vec{r}}{dt} \right|_{\text{INERTIAL}} = \frac{d\vec{R}}{dt} + \left. \frac{d\vec{r}'}{dt} \right|_{\text{BODY}} + \vec{\omega} \times \vec{r}'$$

$$\text{IF RIGID BODY} \quad \left. \frac{d\vec{r}}{dt} \right|_{\text{INERTIAL}} = \frac{d\vec{R}}{dt} + \vec{\omega} \times \vec{r}'$$

For any system of particles:

$$M = \sum m_p$$

$$\vec{R} \equiv \text{CENTER OF MASS} \equiv \frac{\sum m_p \vec{r}_p}{M}$$

BODY

dt INTERNAL dt

For any system of particles:

$$M = \sum_p m_p$$

$$\vec{R} \equiv \text{CENTER OF MASS POSITION} \equiv \frac{\sum_p m_p \vec{r}_p}{\sum_p m_p}$$

$$T = \frac{1}{2} M \dot{\vec{R}}^2 + T'$$

$$T' = \sum_p \frac{m_p}{2} \dot{\vec{r}}_p'^2$$

MEASURED IN FRAME
WITH ORIGIN AT CM
(CM FRAME)

$$\vec{L} = \vec{R} \times M \dot{\vec{R}} + \vec{L}'$$

$$\vec{L}' = \sum_p \vec{r}_p' \times m_p \dot{\vec{r}}_p'$$

For rigid body $\dot{\vec{r}}_p' = \vec{\omega} \times \vec{r}_p'$

$$L = \mathbf{R} \times M \mathbf{R} + L' \quad L' = \sum_p \mathbf{r}_p' \times m_p \dot{\mathbf{r}}_p' \quad \swarrow \quad (\text{CM FRAME})$$

For rigid body $\dot{\mathbf{r}}_p' = \dot{\boldsymbol{\omega}} \times \mathbf{r}_p'$

$$L = \mathbf{R} \times M \dot{\mathbf{R}} + \underbrace{\sum_p \mathbf{r}_p' \times m_p (\dot{\boldsymbol{\omega}} \times \mathbf{r}_p')}_{\text{CALCULATED BEFORE}}$$

$$L' \equiv \sum_j \bar{I}_{ij} \omega_j$$

$$T = \frac{1}{2} M |\dot{\mathbf{R}}|^2 + \underbrace{\sum_p \frac{m_p}{2} (\dot{\boldsymbol{\omega}} \times \mathbf{r}_p')^2}_{\text{CALCULATED BEFORE}}$$

$$T' \equiv \frac{1}{2} \sum_{ij} \bar{I}_{ij} \omega_i \omega_j$$

EULER'S EQUATIONS

$$\frac{d\vec{L}}{dt} = \vec{N} \quad \text{IF} \left\{ \begin{array}{l} \times \text{ INERTIAL FRAME} \\ \times \text{ CM FRAME (ORIGIN AT CM)} \\ \quad \text{MAY BE ACCELERATED BUT} \\ \quad \text{IT STAYS PARALLEL TO INERTIAL FRAME} \end{array} \right\} \Leftrightarrow \text{"TRANSLATED BUT NOT ROTATED"}$$

$$\vec{N} = \frac{d\vec{L}}{dt} = \left. \frac{d\vec{L}}{dt} \right|_{\text{BODY}} + \vec{\omega} \times \vec{L}$$

$$\frac{d}{dt} = \frac{d}{dt} \Big|_{\text{BODY}}$$

$$N_1 = \frac{dL_1}{dt} \Big|_{\text{BODY}} + (\vec{\omega} \times \vec{L})_1 = \frac{dL_1}{dt} \Big|_{\text{BODY}} + \omega_2 L_3 - \omega_3 L_2$$

$$L_i = I_i \omega_i \quad \dot{I}_i \Big|_{\text{BODY}} = 0 \quad (\text{RIGID BODY} \Rightarrow \vec{r}_P = \vec{r} \text{ IN BODY FRAME} \Rightarrow \dot{I}_i \Big|_{\text{BODY}} = 0)$$

$$= I_1 \frac{d\omega_1}{dt} \Big|_{\text{BODY}} + \omega_2 I_3 \omega_3 - \omega_3 I_2 \omega_2 = I_1 \frac{d\omega_1}{dt} \Big|_{\text{BODY}} + \omega_2 \omega_3 (I_3 - I_2)$$

$$\tau = \frac{dL_1}{dt} = \dots + \dots + N_1$$

$$= \left. \frac{dw_1}{dt} \right|_{\text{BODY}} + w_2 \omega_3 \omega_3 - \omega_3 \omega_2 \omega_2 = \left. \frac{dw_1}{dt} \right|_{\text{BODY}} + w_2 \omega_3 (\omega_3 - \omega_2)$$

$$I_1 \left. \frac{dw_1}{dt} \right|_{\text{BODY}} = w_2 \omega_3 (\omega_3 - \omega_2) + N_1$$

$$I_2 \left. \frac{dw_2}{dt} \right|_{\text{BODY}} = w_3 \omega_1 (\omega_1 - \omega_2) + N_2$$

$$I_3 \left. \frac{dw_3}{dt} \right|_{\text{BODY}} = w_1 \omega_2 (\omega_2 - \omega_3) + N_3$$

EULER'S
EQUATIONS

APPLICATIONS

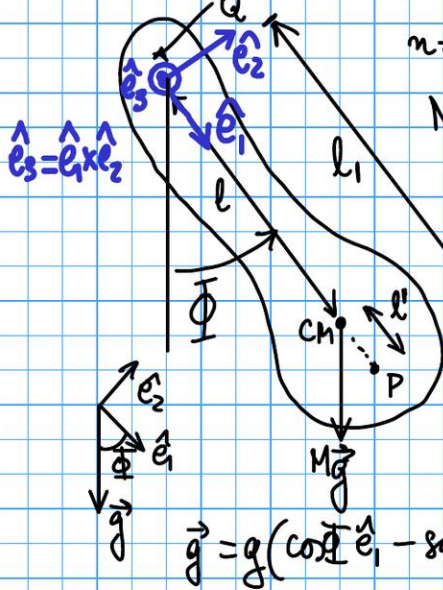


APPLICATIONS

Kater's Pendulum

APPLICATIONS

Kater's Pendulum



$n=1$ degree of freedom $\Leftrightarrow$ angle $\Phi \equiv$ gen coord

NO FRICTION

In Lab frame:

$$I \dot{\omega}_3 = \frac{dL_3}{dt} = N_3 = -Mg l \sin \Phi \quad \omega_3 = \dot{\Phi}$$

$$\vec{N}_g = l \hat{e}_1 \times M \vec{g} = l \hat{e}_1 \times Mg (\cos \Phi \hat{e}_1 - \sin \Phi \hat{e}_2) = -Mg l \sin \Phi \hat{e}_3$$

SMALL ANGLE $\sin \Phi \approx \Phi$

$$I \ddot{\Phi} \approx -Mg l \Phi \quad \ddot{\Phi} \approx -\frac{Mg l}{I} \Phi$$

$$\ddot{\Phi} = -\Omega^2 \Phi \quad \Omega = \left(\frac{Mg l}{I} \right)^{1/2} \text{ FREQ. OF OSCILLATION}$$

Parallel axis theorem

$$I \equiv \bar{I} + M l^2 = M \bar{R}^2 + M l^2 = M (\bar{R}^2 + l^2) = M k^2$$

DEFINE RADIUS OF GYRATION k

By $I \equiv M k^2$

$$I \equiv M k^2$$

DEFINE $k_1 \equiv \frac{k^2}{l} = l + \frac{\bar{R}^2}{l}$

$$I_1 \equiv \frac{k^2}{l}$$

By $I = M k^2$

$$\Omega^2 = \frac{Mgl}{M(k^2 + l^2)} = \frac{g}{l_1}$$

$$[\Omega^2] = \frac{\cancel{L}/T^2}{\cancel{L}} = \frac{1}{T^2} \Rightarrow [\Omega] = \frac{1}{T}$$

COMPARE WITH SIMPLE PENDULUM $\Omega = \sqrt{\frac{g}{l}}$



point mass

$$\bar{I} = 0 \Rightarrow \bar{k} = 0 \quad l_1 = l + \frac{\bar{k}^2}{l} = l$$

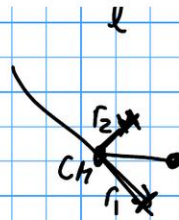
=

(13) (2.7)

PRINT MASS

$$\bar{I} = \int d^3r \, \rho(\vec{r}) (r_1^2 + r_2^2)$$

distances measured wrt CH



POINT MASS

$$r_1 = r_2 = 0$$

$$\bar{k}^2 = \frac{\bar{I}}{M} = \frac{\int d^3r \, \rho(\vec{r}) (r_1^2 + r_2^2)}{\int d^3r \, \rho(\vec{r})}$$

↔ Average squared distance to axis of rotation weighted by the mass density $\rho(\vec{r})$

Suspend the pendulum from point P "Percussion center" at distance l , from Q,

$$M \int d^3r \rho(\vec{r})$$

weighted by the mass density $\rho(\vec{r})$

Suspend the pendulum from point P "Percussion center" at distance l_1 from Q,
along the direction that goes through CM

$$\Omega_Q^2 = \frac{Mgl}{M(\bar{k}^2 + l^2)} = \frac{g}{l_1}$$

$$\Omega_P^2 = \frac{gl_1}{\bar{k}^2 + l_1^2} = \frac{g(l_1 - l)}{\bar{k}^2 + (l_1 - l)^2} = \frac{g \bar{k}^2 l}{\bar{k}^2 + \bar{k}^4/l^2} = \frac{gl}{l^2 + \bar{k}^2} = \Omega_Q^2$$

$$l_1 = l + \frac{\bar{k}^2}{l} \quad l_1 - l = \frac{\bar{k}^2}{l} \quad (l_1 - l)^2 = \frac{\bar{k}^4}{l^2}$$

$$l \quad l \quad l^2$$

Torque - Free Motion in 3D : SYMMETRIC TOP $I_1 = I_2 \neq I_3$ $\Omega \equiv \omega_3 \frac{I_3 - I_1}{I_1}$

$$\begin{aligned} I_1 \frac{d\omega_1}{dt} \Big|_{\text{BODY}} &= \omega_2 \omega_3 (I_2 - I_3) \Rightarrow \dot{\omega}_1 = - \left[\omega_3 \frac{(I_3 - I_2)}{I_1} \right] \omega_2 \Rightarrow \dot{\omega}_1 = -\Omega \omega_2 \\ I_2 \frac{d\omega_2}{dt} \Big|_{\text{BODY}} &= \omega_3 \omega_1 (I_3 - I_1) \Rightarrow \dot{\omega}_2 = \left[\omega_3 \frac{(I_3 - I_1)}{I_2} \right] \omega_1 \Rightarrow \dot{\omega}_2 = \Omega \omega_1 \\ I_3 \frac{d\omega_3}{dt} \Big|_{\text{BODY}} &= \omega_1 \omega_2 (I_1 - I_2) \Rightarrow \dot{\omega}_3 = 0 \end{aligned}$$

$\dot{\vec{\omega}} = \Omega \hat{e}_3 \times \vec{\omega} \Leftrightarrow \vec{\omega}$ rotates with angular velocity $\Omega \hat{e}_3 \Rightarrow$ PRECESSION

$$\dot{\vec{\omega}} = \Omega \hat{e}_3 \times \vec{\omega} = \Omega \hat{e}_3 \times (\omega_1 \hat{e}_1 + \omega_2 \hat{e}_2 + \omega_3 \hat{e}_3) = \Omega (\omega_1 \hat{e}_2 - \omega_2 \hat{e}_1)$$

$$\dot{\omega}_1 = -\Omega \omega_2 \quad \dot{\omega}_2 = \Omega \omega_1 \quad \dot{\omega}_3 = 0$$

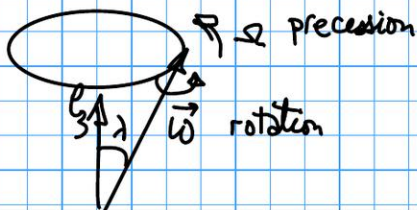
$\vec{\omega}$ rotates around $\hat{e}_3$ with angular velocity Ω

$$\omega_1 = -\Omega \omega_2 \quad \omega_2 = \Omega \omega_1 \quad \omega_3 = 0$$

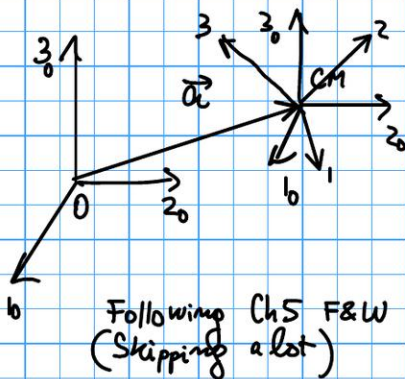
$\vec{\omega}$ rotates around $\hat{e}_3$ with ang speed Ω

If initially $\vec{\omega} = \omega (\sin \lambda \hat{e}_1 + \cos \lambda \hat{e}_3)$ $\vec{\omega} \in$ plane of $\hat{e}_1$ and $\hat{e}_3$

$$\omega_1 = \omega \sin \lambda \cos(\Omega t) \quad \omega_2 = \omega \sin \lambda \sin(\Omega t) \quad \omega_3 = \omega \cos \lambda$$



$\hat{e}_1^0 \hat{e}_2^0 \hat{e}_3^0$ INERTIAL SET OF UNIT VECTORS



$\hat{e}_1^0, \hat{e}_2^0, \hat{e}_3^0$ INERTIAL SET OF UNIT VECTORS
(NOT ROTATING)

$\hat{e}_1, \hat{e}_2, \hat{e}_3$ BODY SET OF UNIT VECTORS
(ROTATING WITH BODY)

We need 6 coordinates to describe positions of all points
in a RIGID BODY

CM POSITION $\vec{R} \leftrightarrow X, Y, Z$

3 coord to fix orientation of
 $\hat{e}_1, \hat{e}_2, \hat{e}_3$ w.r.t $\hat{e}_1^0, \hat{e}_2^0, \hat{e}_3^0$

WE CHOOSE EULER ANGLES: α, β, γ

- $\hat{e}_3 \perp \hat{e}_1^0, \hat{e}_2^0$
- ORIENTATION OF NEW NORTH POLE $\rightarrow \beta$ POLAR
 $\rightarrow \alpha$ AZIMUTHAL
- γ SPECIFIES ADDITIONAL ROTATION AROUND $\hat{e}_3$
- α, β, γ CAN BE CHOSEN INDEPENDENTLY AND THEY COMPLETELY
SPECIFY ORIENTATION OF BODY FRAME $\Rightarrow$ USE α, β, γ
AS GENERALIZED COORD

$$T = \frac{1}{2} (\mathcal{I}_1 \omega_1^2 + \mathcal{I}_2 \omega_2^2 + \mathcal{I}_3 \omega_3^2) \quad (1)$$

$$\vec{\omega} = \alpha \hat{e}_3^0 + \beta \hat{e}_\beta + \gamma \hat{e}_3 \quad (2) \quad \hat{e}_\beta = \cos \gamma \hat{e}_2 + \sin \gamma \hat{e}_1 \quad (3)$$

$$\hat{e}_3^0 = \cos \beta \hat{e}_3 + \sin \beta [\cos \gamma (-\hat{e}_1) + \sin \gamma \hat{e}_2] \quad (4)$$

$$\vec{\omega} = (-\alpha \sin \beta \cos \gamma + \beta \sin \gamma) \hat{e}_1 + (\alpha \sin \beta \sin \gamma + \beta \cos \gamma) \hat{e}_2 + (\gamma + \alpha \cos \beta) \hat{e}_3 \quad (5)$$

NOTE:

Infinitesimal rot

$$R_k = 1 + M_k \dot{\alpha} dt \quad (6)$$

$$R_\beta = 1 + M_\beta \dot{\beta} dt \quad (7)$$

$$R_\alpha R_\beta = 1 + (M_\alpha \dot{\alpha} + M_\beta \dot{\beta}) dt + \cancel{O(dt^2)} \quad (8)$$

$$R_\beta R_\alpha = 1 + (M_\beta \dot{\beta} + M_\alpha \dot{\alpha}) dt + \cancel{O(dt^2)} \quad (9)$$

EQUAL TO ORDER dt

$$\Rightarrow \text{INFITESIMAL ROTATIONS ADD, } \vec{\omega} = \vec{\omega}_x + \vec{\omega}_y + \vec{\omega}_z \quad (10) \Rightarrow (2)$$

KINETIC ENERGY

→ simplifying further, we get 0. (14) (15)

KINETIC ENERGY

General case:

$$T = \frac{1}{2} I_1 \dot{\alpha}^2 + \frac{1}{2} I_2 \dot{\beta}^2 + \frac{1}{2} I_3 \dot{\gamma}^2 = \frac{1}{2} I_1 (-\dot{\alpha} \sin \alpha \cos \alpha + \dot{\beta} \sin \alpha)^2 + \frac{1}{2} I_2 (\dot{\alpha} \sin \alpha \sin \alpha + \dot{\gamma} \cos \alpha)^2 + \frac{1}{2} I_3 (\dot{\gamma} + \dot{\alpha} \cos \alpha)^2 \quad (11)$$

General case:

$$T = \frac{I_1}{2} \omega_1^2 + \frac{I_2}{2} \omega_2^2 + \frac{I_3}{2} \omega_3^2 = \frac{I_1}{2} (-\dot{\alpha} \sin \beta \cos \gamma + \dot{\beta} \sin \gamma)^2 + \frac{I_2}{2} (\dot{\alpha} \sin \beta \sin \gamma + \dot{\beta} \cos \gamma)^2 + \frac{I_3}{2} (\dot{\gamma} + \dot{\alpha} \cos \beta)^2 \quad (11)$$

Symmetric Top:

$$I_1 = I_2 \quad (12)$$

$$T = \frac{I_1}{2} (\dot{\beta}^2 + \dot{\alpha}^2 \sin^2 \beta) + \frac{I_3}{2} (\dot{\gamma} + \dot{\alpha} \cos \beta)^2 \quad (13)$$

Torque-Free Motion of symmetric top

$$L = T = \frac{I_1}{2} (\dot{\beta}^2 + \dot{\alpha}^2 \sin^2 \beta) + \frac{I_3}{2} (\dot{\gamma} + \dot{\alpha} \cos \beta)^2 \quad (14)$$

α, β, γ Gen Coord $\Rightarrow$ 3 Lagrange Eqs

2 CYCLIC COORDINATES: α, γ (15)

$$\Rightarrow \begin{matrix} 2 \text{ CONST} \\ \text{OF} \\ \text{MOTION} \end{matrix} \left\{ \begin{array}{l} p_\alpha = \frac{\partial L}{\partial \dot{\alpha}} = I_1 \sin^2 \beta \dot{\alpha} + I_3 (\dot{\gamma} + \dot{\alpha} \cos \beta) \cos \beta \quad (16) \\ p_\gamma = \frac{\partial L}{\partial \dot{\gamma}} = I_3 (\dot{\gamma} + \dot{\alpha} \cos \beta) \quad (17) \end{array} \right.$$

WE WILL SHOW LATER THAT:

$$p_\alpha = L^z \cdot \hat{e}_3^0 \quad (18)$$

$$p_\gamma = L^z \cdot \hat{e}_3 \quad (19)$$

recall $\Rightarrow$ (5)

$$\text{recall } \vec{\omega} = (-\dot{\alpha} \sin \beta \cos \gamma + \dot{\beta} \sin \gamma) \hat{e}_1 + (\dot{\alpha} \sin \beta \sin \gamma + \dot{\beta} \cos \gamma) \hat{e}_2 + (\dot{\gamma} + \dot{\alpha} \cos \beta) \hat{e}_3 \quad (5)$$

We compute now $p_\gamma = \frac{\partial L}{\partial \dot{\gamma}} = I_3 (\dot{\gamma} + \dot{\alpha} \cos \beta)$ and notice that $p_\gamma \stackrel{(5)}{=} I_3 \omega_3 \stackrel{(19)}{=} \vec{L} \cdot \hat{e}_3$ ✓ AS ANTICIPATED

Also $p_\alpha = \frac{\partial L}{\partial \dot{\alpha}} = I_1 \sin^2 \beta \dot{\alpha} + I_3 (\dot{\gamma} + \dot{\alpha} \cos \beta) \cos \beta$ (16) $\vec{L} = I_1 (\omega_1 \hat{e}_1 + \omega_2 \hat{e}_2) + I_3 \omega_3 \hat{e}_3$ (20)

Compare with $\vec{L} \cdot \hat{e}_3 \stackrel{(20)}{=} (I_1 (\omega_1 \hat{e}_1 + \omega_2 \hat{e}_2) + I_3 \omega_3 \hat{e}_3) \cdot (\cos \beta \hat{e}_3 + \sin \beta [\cos \gamma (-\hat{e}_1) + \sin \gamma \hat{e}_2])$ (21)

$$= I_1 [(-\dot{\alpha} \sin \beta \cos \gamma + \dot{\beta} \sin \gamma)(-\sin \beta \cos \gamma) + (\dot{\alpha} \sin \beta \sin \gamma + \dot{\beta} \cos \gamma)(\sin \beta \sin \gamma)] + I_3 (\dot{\gamma} + \dot{\alpha} \cos \beta) \cos \beta \quad (22)$$

$$= I_1 \dot{\alpha} \sin^2 \beta (\cos^2 \gamma + \sin^2 \gamma) + I_3 (\dot{\gamma} + \dot{\alpha} \cos \beta) \cos \beta \quad (23)$$

$$+ I_3 (\dot{\gamma} + \alpha \cos \beta) \cos \beta \quad (22)$$

$$= I_1 \dot{\alpha} \sin^2 \beta (\omega^2 \gamma + \sin^2 \gamma) + I_3 (\dot{\gamma} + \alpha \cos \beta) \cos \beta \quad (23)$$

$$\boxed{\vec{L} \cdot \hat{e}_3 = I_1 \dot{\alpha} \sin^2 \beta + I_3 (\dot{\gamma} + \alpha \cos \beta) \cos \beta = p_\alpha} \quad \checkmark \quad (18)$$

$$p_\beta = \frac{\partial L}{\partial \dot{\beta}} = I_1 \dot{\beta} \quad (24) \quad \text{NOT A CONSTANT (IN PRINCIPLE) BECAUSE } \beta \text{ APPEARS EXPLICITLY IN THE LAGRANGIAN}$$

$$\frac{\partial L}{\partial \beta} = I_1 \sin \beta \cos \beta \dot{\alpha}^2 - I_3 (\dot{\gamma} + \dot{\alpha} \cos \beta) \dot{\alpha} \sin \beta = \dot{\alpha} \sin \beta (I_1 \dot{\alpha} \cos \beta - I_3 \omega_3) \quad (25)$$

Then

$$\dot{p}_\beta = \frac{\partial L}{\partial \beta} = \dot{\alpha} \sin \beta (I_1 \dot{\alpha} \cos \beta - I_3 \omega_3) \quad (26) \quad (\text{However, see later})$$

Compare p_β with $\vec{L} \cdot \hat{e}_\beta = \overset{(20)}{\rightarrow} (I_1 (\omega_1 \hat{e}_1 + \omega_2 \hat{e}_2) + I_3 \omega_3 \hat{e}_3) \overset{(3)}{\rightarrow} (\cos \gamma \hat{e}_2 + \sin \gamma \hat{e}_1)$

Recall $\vec{\omega} = (-\dot{\alpha} \sin \beta \cos \gamma + \dot{\beta} \sin \gamma) \hat{e}_1 + (\dot{\alpha} \sin \beta \sin \gamma + \dot{\beta} \cos \gamma) \hat{e}_2 + (\dot{\gamma} + \dot{\alpha} \cos \beta) \hat{e}_3 \quad (5)$

Compare p_β with $\vec{L} \cdot \hat{e}_\beta = (I_1 (\omega_1 \hat{e}_1 + \omega_2 \hat{e}_2) + I_3 \omega_3 \hat{e}_3) (\cos \gamma \hat{e}_2 + \sin \gamma \hat{e}_1)$

Recall $\vec{\omega} = (-\dot{\alpha} \sin \beta \cos \gamma + \dot{\beta} \sin \gamma) \hat{e}_1 + (\dot{\alpha} \sin \beta \sin \gamma + \dot{\beta} \cos \gamma) \hat{e}_2 + (\dot{\gamma} + \dot{\alpha} \cos \beta) \hat{e}_3$ (5)

$$\vec{L} \cdot \hat{e}_\beta = I_1 [(-\dot{\alpha} \sin \beta \cos \gamma + \dot{\beta} \sin \gamma) \sin \gamma + (\dot{\alpha} \sin \beta \sin \gamma + \dot{\beta} \cos \gamma) \cos \gamma] \quad (27)$$

$$= I_1 \dot{\beta} (\sin^2 \gamma + \cos^2 \gamma) = I_1 \dot{\beta} \Rightarrow \boxed{\vec{L} \cdot \hat{e}_\beta = p_\beta} \quad (29) \quad \checkmark$$

Description in Inertial (or CM) Frame

No torque $\Rightarrow \dot{\vec{L}} = \vec{0}$ $\Rightarrow$ CHOOSE $\hat{e}_3^0 \parallel \vec{L}^{(32)} \Rightarrow \vec{L} \cdot \hat{e}_3^0 = |\vec{L}|^{(34)}$ AND $\vec{L} = |\vec{L}| \hat{e}_3^0$ (35)

(30) $\Rightarrow$ (31) $\Rightarrow$ (33) $\Rightarrow$ (34)

$|\vec{L}|$ doesn't change (33)

(35) (36) $\rightarrow |L|$ doesn't change (37)

We know that $\vec{L} \cdot \hat{e}_3 = p_\beta$ is a constant of the motion

$$\vec{L} \cdot \hat{e}_3 = \underbrace{|\vec{L}|}_{=p_\beta} \underbrace{\hat{e}_3^0 \cdot \hat{e}_3}_{=const} \Rightarrow \hat{e}_3^0 \cdot \hat{e}_3 \text{ CONSTANT} \quad (37)$$

$const = \hat{e}_3^0 \cdot \hat{e}_3 = \cos(\text{angle between } \hat{e}_3 \text{ and } \hat{e}_3^0) = \cos \beta \quad (38)$

β IS A CONSTANT OF THE MOTION $\Rightarrow \dot{\beta} = 0 \Rightarrow$ (39) (40)

$$p_\beta = I_1 \dot{\beta} = \vec{L} \cdot \hat{e}_\beta = 0 \quad (41)$$

For this particular choice of axes, p_β and $\vec{L} \cdot \hat{e}_\beta$ are zero!!

$$\begin{aligned} (41) \quad \vec{L} \cdot \hat{e}_\beta = 0 \\ \vec{L} \neq 0 \quad |\hat{e}_\beta| = 1 \neq 0 \end{aligned} \Rightarrow \vec{L} \perp \hat{e}_\beta \Rightarrow \vec{L} \text{ is in the plane of } \hat{e}_3 \text{ and } \hat{e}_3^0 \quad (42) \quad (43)$$

CONSTANTS: $I_1, I_3, p_\beta, I_3 \omega_3 = L_3, p_\phi$

$$L \neq 0 \quad |\mathbf{p}| \neq 0 \quad (42)$$

$$v \quad (43)$$

CONSTANTS: $I_1, I_3, \beta, I_3 \omega_3 = L_3, p_\alpha$

$$p_\alpha = \frac{\partial L}{\partial \dot{\alpha}} = I_1 \sin^2 \beta \dot{\alpha} + I_3 (\dot{\gamma} + \dot{\alpha} \cos \beta) \cos \beta \quad (16) \quad (5) \Rightarrow \underline{I_1 \sin^2 \beta \dot{\alpha} + I_3 \omega_3 \cos \beta} \quad (44) \Rightarrow \boxed{\dot{\alpha} \text{ CONSTANT}} \quad (45)$$

$$p_\gamma = \frac{\partial L}{\partial \dot{\gamma}} = I_3 (\dot{\gamma} + \dot{\alpha} \cos \beta) \quad (17) \Rightarrow \boxed{\dot{\gamma} \text{ CONSTANT}} \quad (46)$$

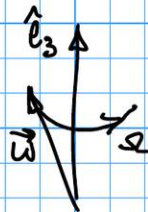
$I_3, p_\alpha, \dot{\alpha}, \beta \text{ CONSTANT}$

$$\alpha = \dot{\alpha}_0 t \quad (47) \uparrow \text{CONSTANTS}$$

$$\gamma = \dot{\gamma}_0 t \quad (48) \uparrow \text{CONSTANTS}$$

$$(4) \Rightarrow 0 = \dot{p}_\beta = \frac{\partial L}{\partial \beta} = \dot{\alpha} \sin \beta (I_1 \dot{\alpha} \cos \beta - I_3 \omega_3) \Rightarrow \boxed{\dot{\alpha} \cos \beta = \frac{I_3}{I_1} \omega_3} \quad (49)$$

$$(5) \Rightarrow \omega_3 = \dot{\gamma} + \dot{\alpha} \cos \beta = \dot{\gamma} + \frac{I_3}{I_1} \omega_3 \Rightarrow \boxed{\dot{\gamma} = \omega_3 \frac{I_1 - I_3}{I_1} = -\Omega} \quad (51)$$



(50) $\Omega = \omega_3 \frac{I_3 - I_1}{I_1}$ defined when looking at body frame evolution of $\vec{\omega}$ a couple of lectures before

$$(2) \Rightarrow (40) \Rightarrow \boxed{\vec{\omega} = \dot{\alpha} \hat{e}_3^0 + \dot{\gamma} \hat{e}_3} \quad (52) \Rightarrow \boxed{\hat{e}_3^0, \hat{e}_3, \vec{\omega}, \vec{e}} \text{ are in one plane} \quad (53)$$

Consider cases where $I_1 = I_2 < I_3$ or $I_1 = I_2 > I_3$

$\vec{e} = \frac{1}{\omega} (\omega_1 \hat{e}_1 + \omega_2 \hat{e}_2) \perp \omega \hat{e}_3$

Consider cases where $I_1 = I_2 < I_3$ or $I_1 = I_2 > I_3$

$$\vec{L} = I_1 (\omega_1 \hat{e}_1 + \omega_2 \hat{e}_2) + I_3 \omega_3 \hat{e}_3 \quad (20)$$

SIMPLE PENDULUM:

$$\ddot{\theta} = -\frac{g}{l} \sin \theta$$

THUMBTRACK:

$$(\text{angle})'' = - \left[\begin{array}{c} \text{A.WFUL} \\ \text{MESS} \end{array} \right] \sin(\text{angle})$$
